BASIC INTERNET RESEARCH TUTORIAL

Department Information and Communication Technology
Mzuzu University

The Tutorial will cover

- Internet Research

Core components

- Internet
- Browsers
- Search Engines
- Subject Directories
- Websites

What is the Internet ?

A network made up of hundred of thousand of computers all over the world, with millions of users connected electronically

What can you do with the Internet ?

- Information Browsing
- File Transfer
- Use of other computers
- International debates or discussion groups
- Electronic mail

Computers on the internet can use one or all of these:

- Electronic mail
- Telnet
- FTP
- The WWW

Electronic Mail

Permits you to send and receive mail and provides access to discussion groups often called listserv.

Telnet

Is a program that lets you use power of the Internet to connect you to databases, library catalogs and other information resources around the world.

Permits your computer to act as if it were directly connected to another computer.

File Transfer Protocol

Allows your computer to rapidly retrieve complex files intact from a remote computer and view or save them on your computer

The WWW

The largest, fastest growing activity on the internet.

It incorporates all of the internet. You can retrieve documents, view images, animation, video, sound, speak and hear voice.

The WWW

- The Web is the part or service of the Internet that uses text, images, sound, and video.
- It links documents together
- This linking is sometimes called hypertext.

The WWW

- The **World Wide Web** is a collection of electronic documents that are linked together like a spider web.
- These documents are stored on computers called **servers** located around the world

What is a Web Browser ?

A Web browser contains the basic software you need in order to find, retrieve, view, and send information over the Internet.

This includes software that lets you browse – or search - the World Wide Web.

Types of browsers

- **Graphical**: Text, images, audio, and video are retrievable through a graphical software program such as Internet Explorer, Netscape Navigator
- **Text**: Lynx is a browser that provides access to the Web in text-only mode

Examples of Web Browsers

- Internet Explorer



- Mozilla Firefox



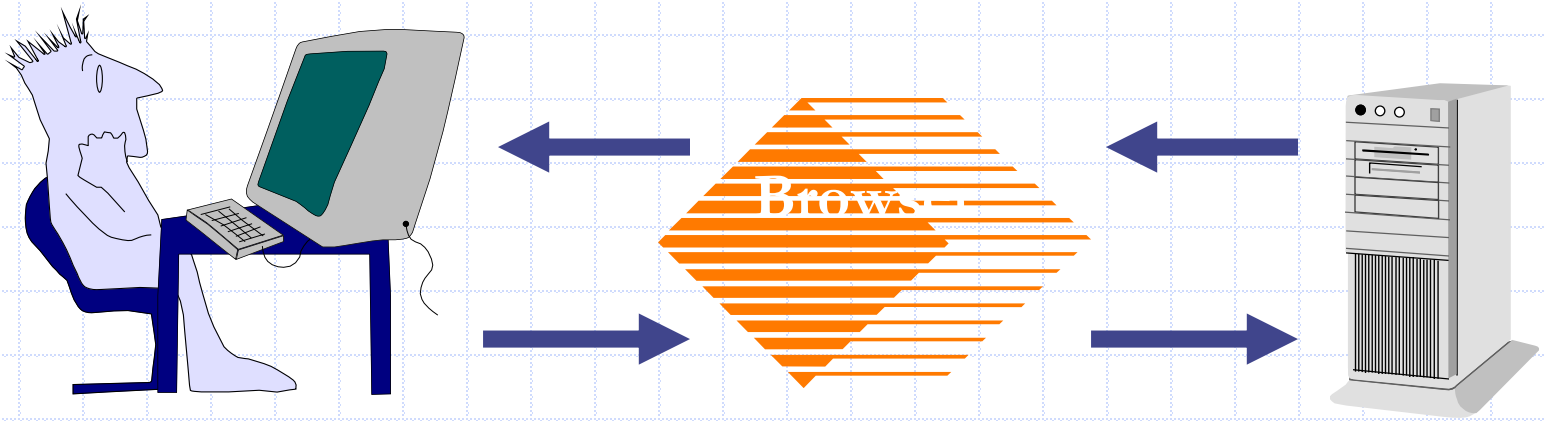
- Netscape



How the Web Works

5. User receives file displayed by the browser

4. Server sends requested files to browser to be interpreted



1.User sends request

2. Browser interprets user's selection and makes request from appropriate server

3. Server accepts and processes request from browser

What are Search Engines ?

A search engine is a service that indexes, organizes, and often rates and reviews Web sites.

Search Engine

- For targeted, complex, and sometimes general queries, use a search engine
- A search engine is a class of computer software designed specifically to search World Wide Web documents for keywords entered by a searcher.

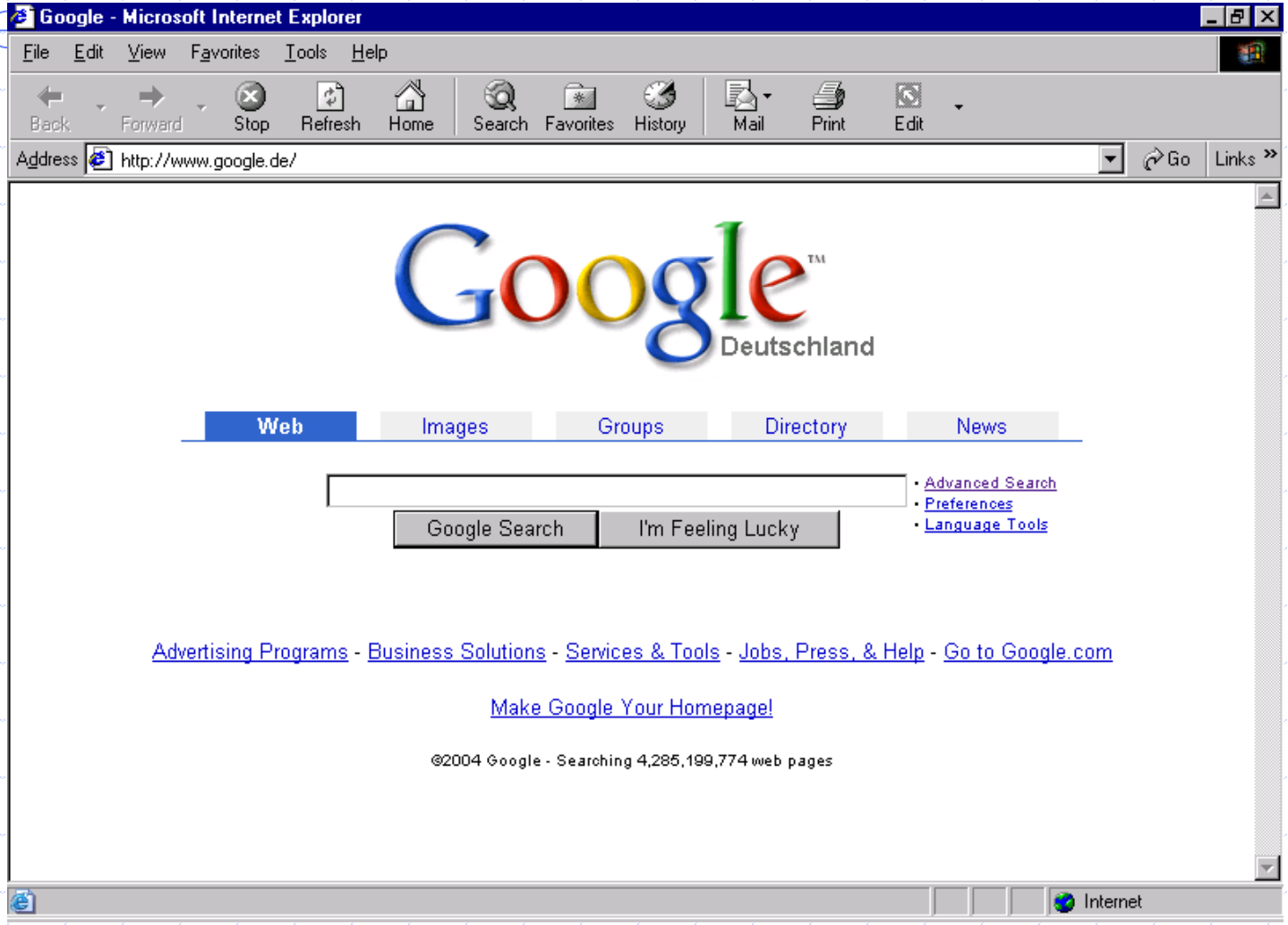
Search Engine

- The search engine retrieves a list of documents containing the specified keywords. Well known examples of search engines include Web Crawler, Ask Jeeves and Google.

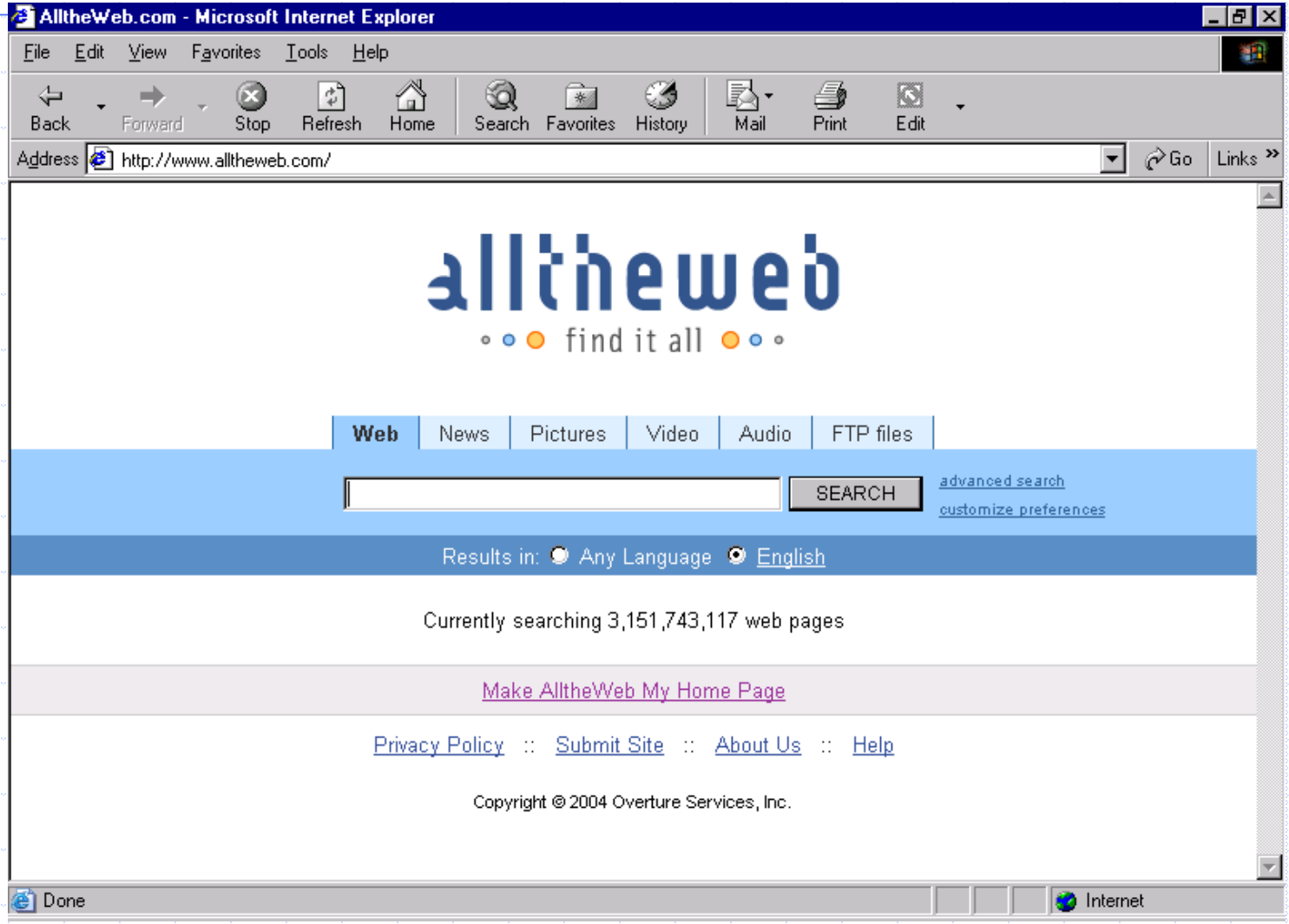
Search Engine

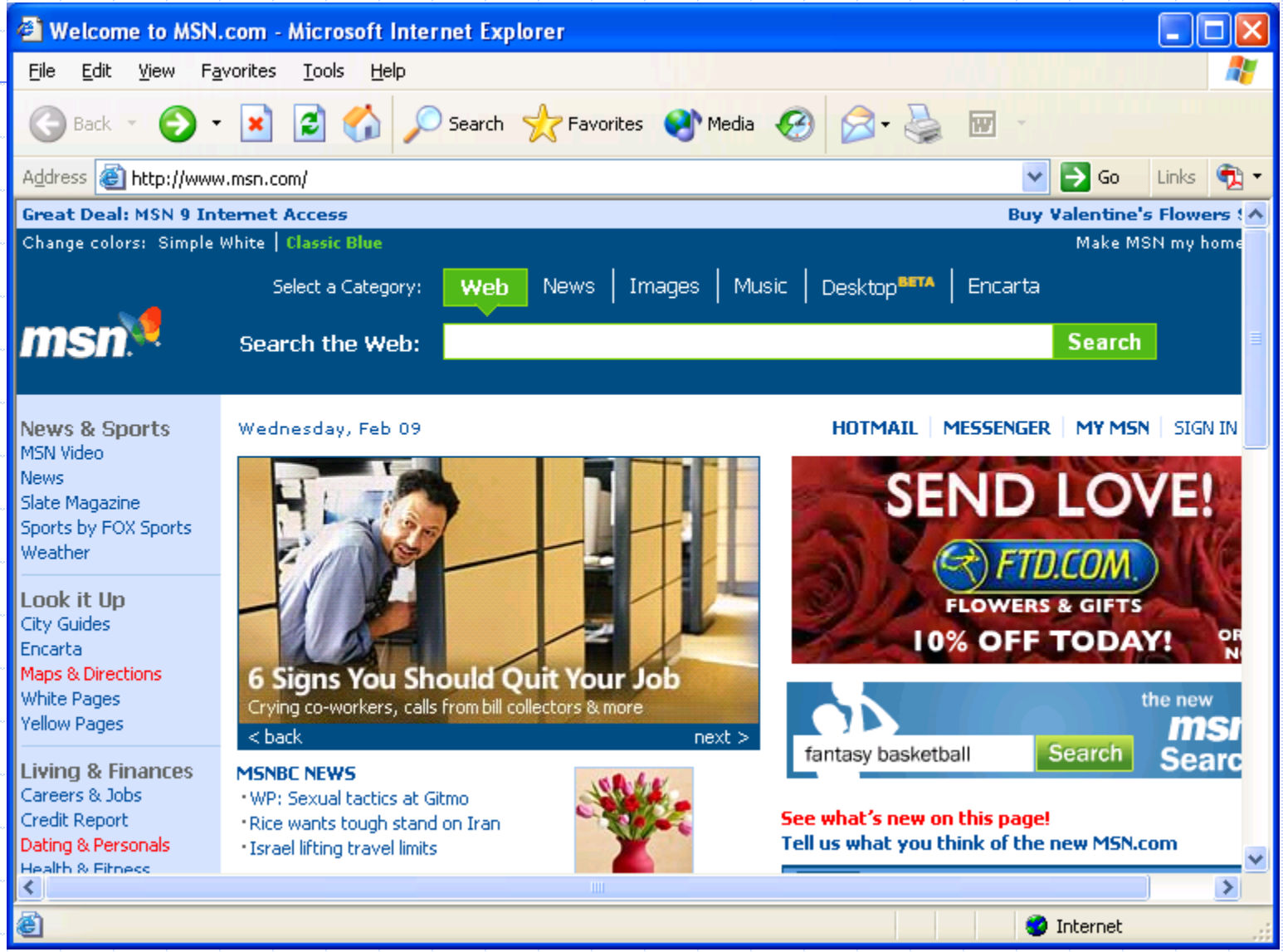
- A search engine is a searchable database of Internet files collected by a computer program (called a wanderer, crawler, robot, worm, spider).
- Indexing is created from the collected files, e.g., title, full text, size, URL, etc

http://www.google.com



http://www.alltheweb.com




Welcome to MSN.com - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Search Favorites Media
Address http://www.msn.com/
Go Links
Great Deal: MSN 9 Internet Access
Buy Valentine's Flowers
Change colors: Simple White | Classic Blue
Make MSN my home
Select a Category: Web News Images Music Desktop BETA Encarta
msn
Search the Web:
Search
News & Sports
MSN Video
News
Slate Magazine
Sports by FOX Sports
Weather
Look it Up
City Guides
Encarta
Maps & Directions
White Pages
Yellow Pages
Living & Finances
Careers & Jobs
Credit Report
Dating & Personals
Health & Fitness
Wednesday, Feb 09
HOTMAIL MESSENGER MY MSN SIGN IN
6 Signs You Should Quit Your Job
Crying co-workers, calls from bill collectors & more
< back
next >
SEND LOVE!
FTD.COM
FLOWERS & GIFTS
10% OFF TODAY!
the new
fantasy basketball
Search
MSNBC NEWS
• WP: Sexual tactics at Gitmo
• Rice wants tough stand on Iran
• Israel lifting travel limits
See what's new on this page!
Tell us what you think of the new MSN.com
Internet

Subject Directory

- **Definition:** A subject directory is a service that offers a collection of links to Internet resources submitted by site creators or evaluators and organized into subject categories

Subject Directories

These can browse subject categories or search using broad, general terms. Searches need to be less specific than search engines

Subject Directory

- For general, research oriented queries, and when you want to view sites often recommended by experts, use a subject directory

Subject Directory

- Directory services use selection criteria for choosing links to include, though the selectivity varies among services
- Most directories are searchable

Subject Directory

- There are two basic types of directories:
  - *academic and*
  - *professional directories*
- often created and maintained by subject experts to support the needs of researchers, and

Subject Directory

- directories contained on *commercial portals* that cater to the general public and are competing for traffic

Subject Directory

- INFOMINE (http://infomine.ucr.edu/), from the University of California, is a good example of an academic subject directory.

Welcome to INFOMINE: Scholarly Internet Resource Collections - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Search Favorites Media

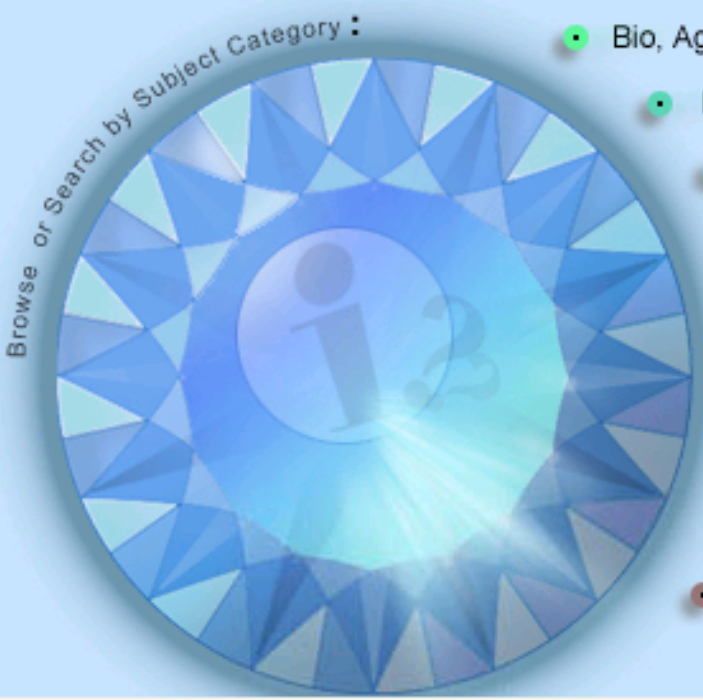
Address http://infomine.ucr.edu/ Go Links

124,054 Academically Valuable Resources

Search for: GO

Advanced Search | Search Tips

Celebrating Ten Years of Service!



- Bio, Ag & Med Sciences
- Business & Economics
- Cultural Diversity
- Ejournals
- Government Info
- Maps & GIS
- PhysSci, Engr, CS & Math
- SocSci & Humanities
- Visual & Performing Arts

Internet

Librarians' Index to the Internet - http://lii.org/



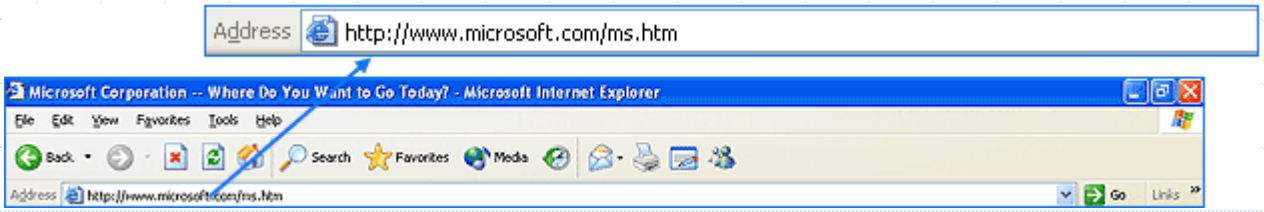
Subject Directory

- Yahoo! (http://www.yahoo.com) is a good example of a commercial portal--but it should never be used for serious research.

YAHOO!


Yahoo! - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Search Favorites Media
Address http://www.yahoo.com/
Go Links
Make Yahoo! your home page
Get Yahoo! Toolbar with Anti-S
Finance Music Shopping YAHOO! Mail My Yahoo! Messenger
Select Search Category: Web Images Video NEW Directory Local News Products
Search the Web:
Yahoo! Search
Advanced
Preferences
Yahoo! Search - Popular terms: Paris Hilton, Valentine's Day, American Idol, IRS, Prom Dresses
Free mail Sign Up
Mail status: Sign In
Autos Chat Finance Games GeoCities Groups Health
Horoscopes HotJobs Kids Mail Maps Messenger Mobile
Movies Music My Yahoo! News People Search Personals Photos
Real Estate Shopping Sports Travel TV Yellow Pages
All Y! Services...
$644 $349 Desktop
One Day Only!
with 17" CRT Monitor AND CD Burner!
Expires 02/10/05.
After $150 mail-in rebate and $145 instant savings.
Small Business
More Details
$349 Dell Desktop - Ad Feedback
In the News
1:38am, Thu Feb
Rice: Iran risks U.N. action on nuke plan
Medicare drug benefit to cost $724 billion
Experts urge routine HIV tests for all
Senate refuses change to class-action bill
Yahoo! Shopping - Last-Minute Gift Center
Special Offer
Flowers
Electronics
Jewelry
Love Gifts
Internet

What is a URL ?



A URL (or uniform resource locator) is the address of an Internet page on a Web site. E.g. http://usembassy.state.gov/malawi

THE URL

- The URL specifies the Internet address of a file stored on a host computer connected to the Internet.
- Every file on the Internet, has a unique URL.

THE URL

- Web software programs use the URL to retrieve the file from the host computer and the directory in which it resides
- This file is then displayed on the monitor connected to the user's local machine

Anatomy of a URL

- This is the format of the URL:
- protocol://host/path/filename
  - For example:
  - http://www.house.gov/agriculture/schedule.htm

Anatomy of a URL

Other examples:

- **telnet://opac.albany.edu** the University at Albany library text-based catalog
- **ftp://ftp.uu.net/graphics/picasso** a file at an ftp site

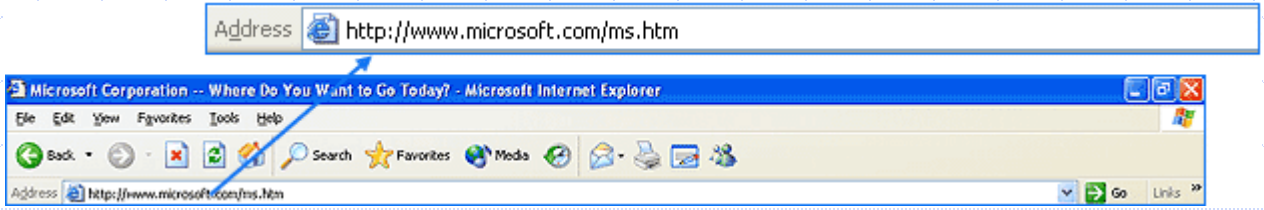
What is a website?

- This is a collection of WebPages
- Mainly connected through the use of hyperlinks
- Has a common URL

How do you search for info on the internet?

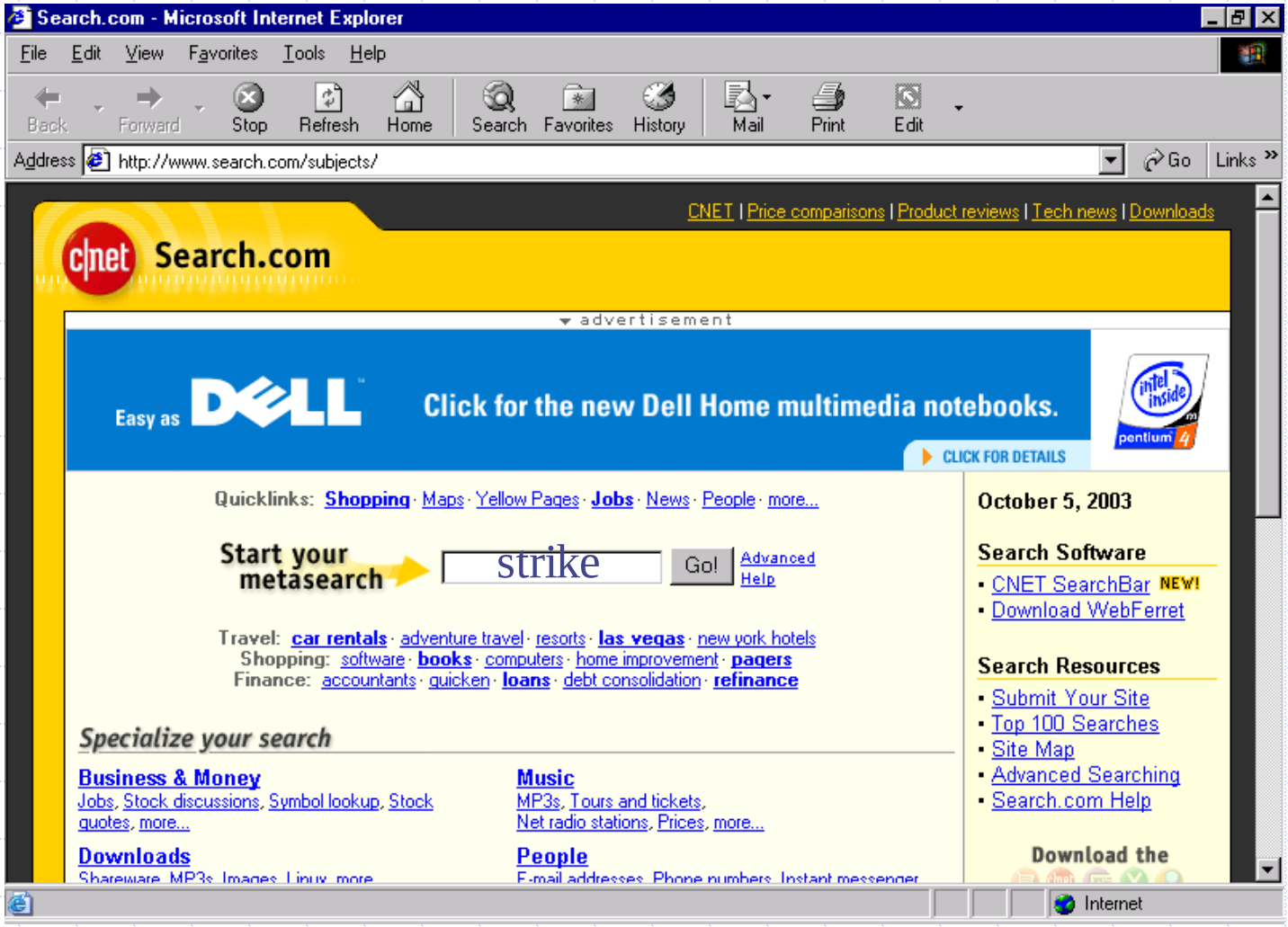
- Using the address bar
- Using free text
- Using Boolean operators
- Phrase searching
- Using the Advanced Search Option

Searching using the address bar



Write the website address(URL) in the address bar.

Searching using free text



Phrase searching

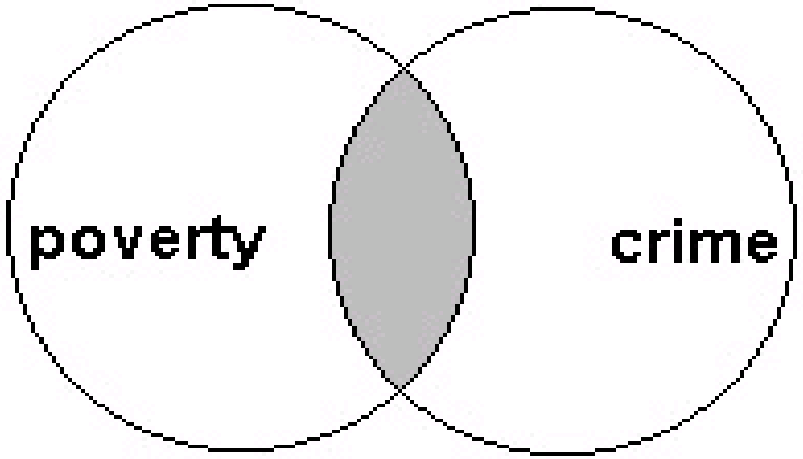
Requires all your terms to appear in exactly the order you enter them. Enclose the term in double quotations e.g., “instructional design ”

Searching using Boolean Operators

- AND
- OR
- NOT

The **AND** operator

AND finds pages containing all your words

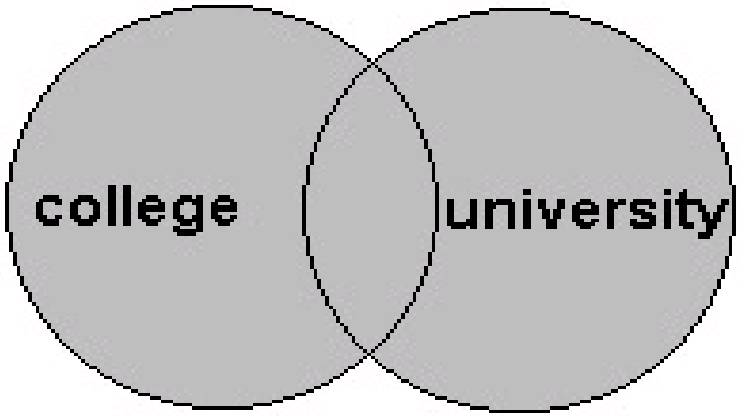


The **OR** operator

- OR finds pages containing any of your words

nuclear OR atomic

bomb (nuclear OR atomic)

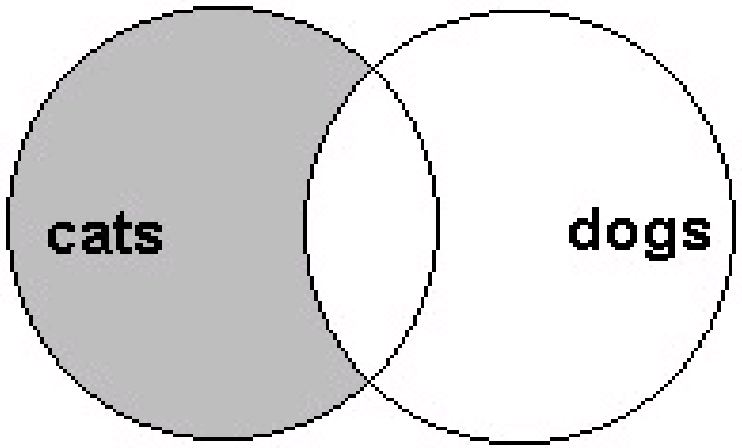


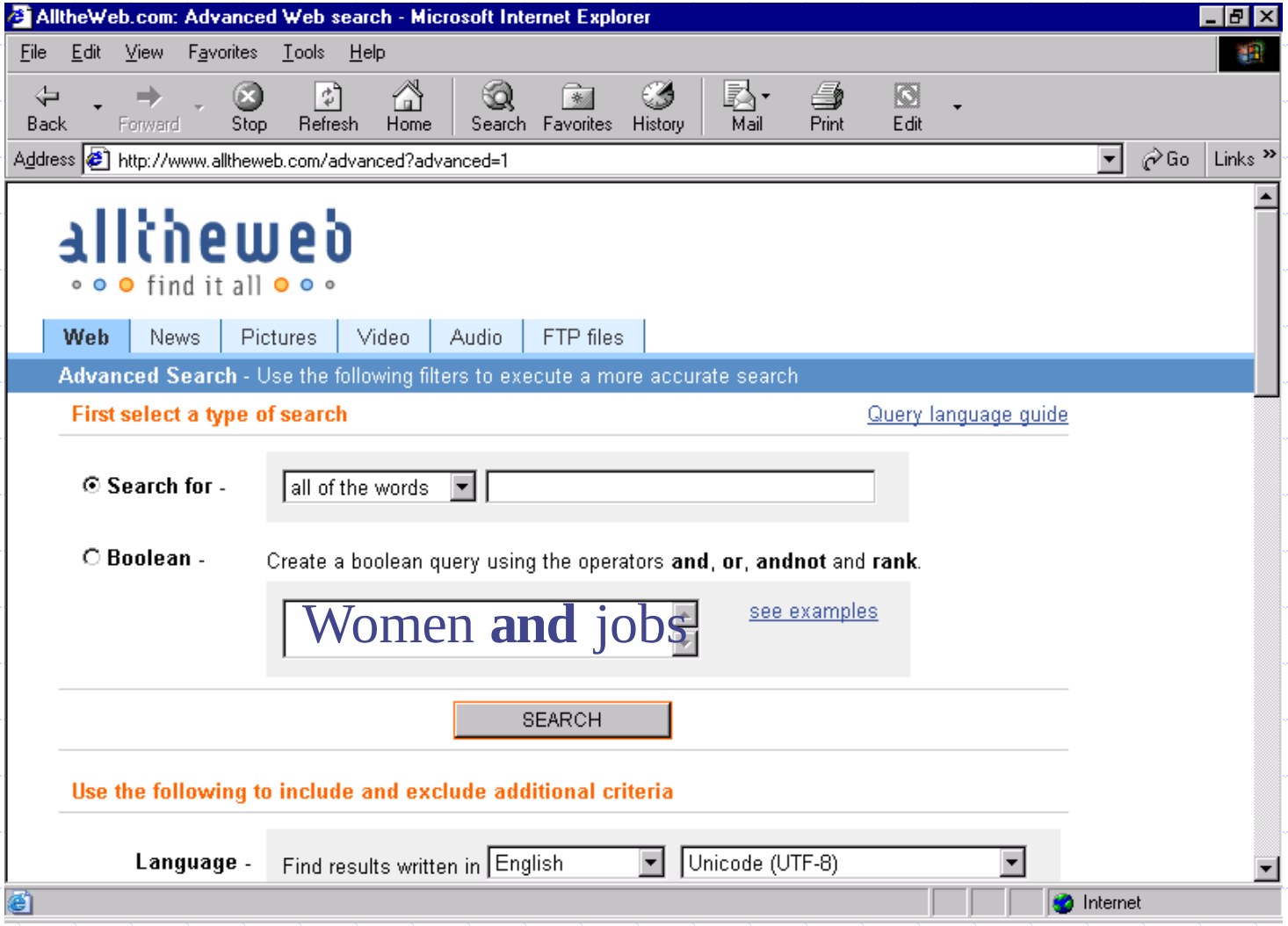
The **NOT** operator

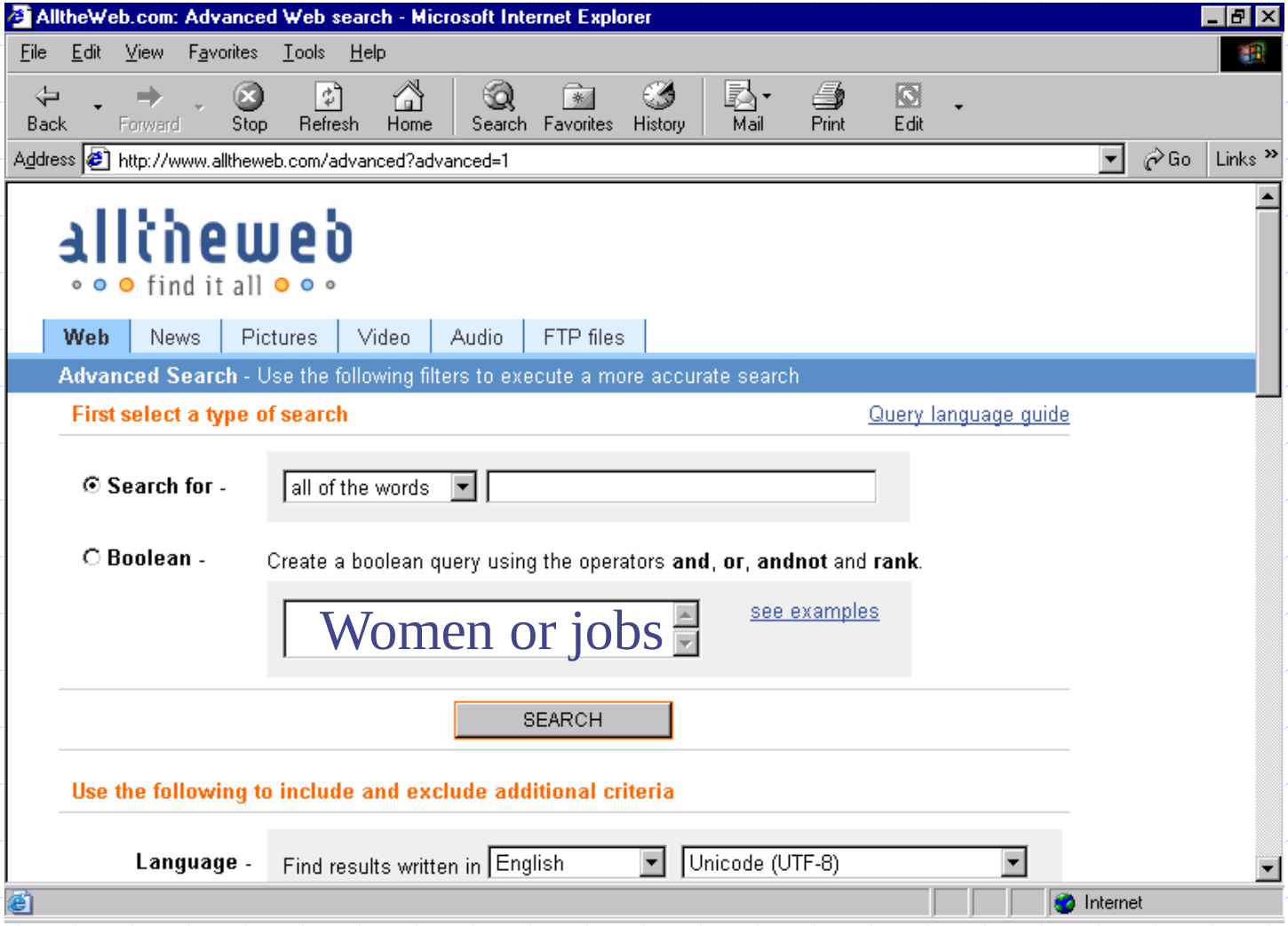
- NOT or - excludes any page containing that word (use cautiously!)

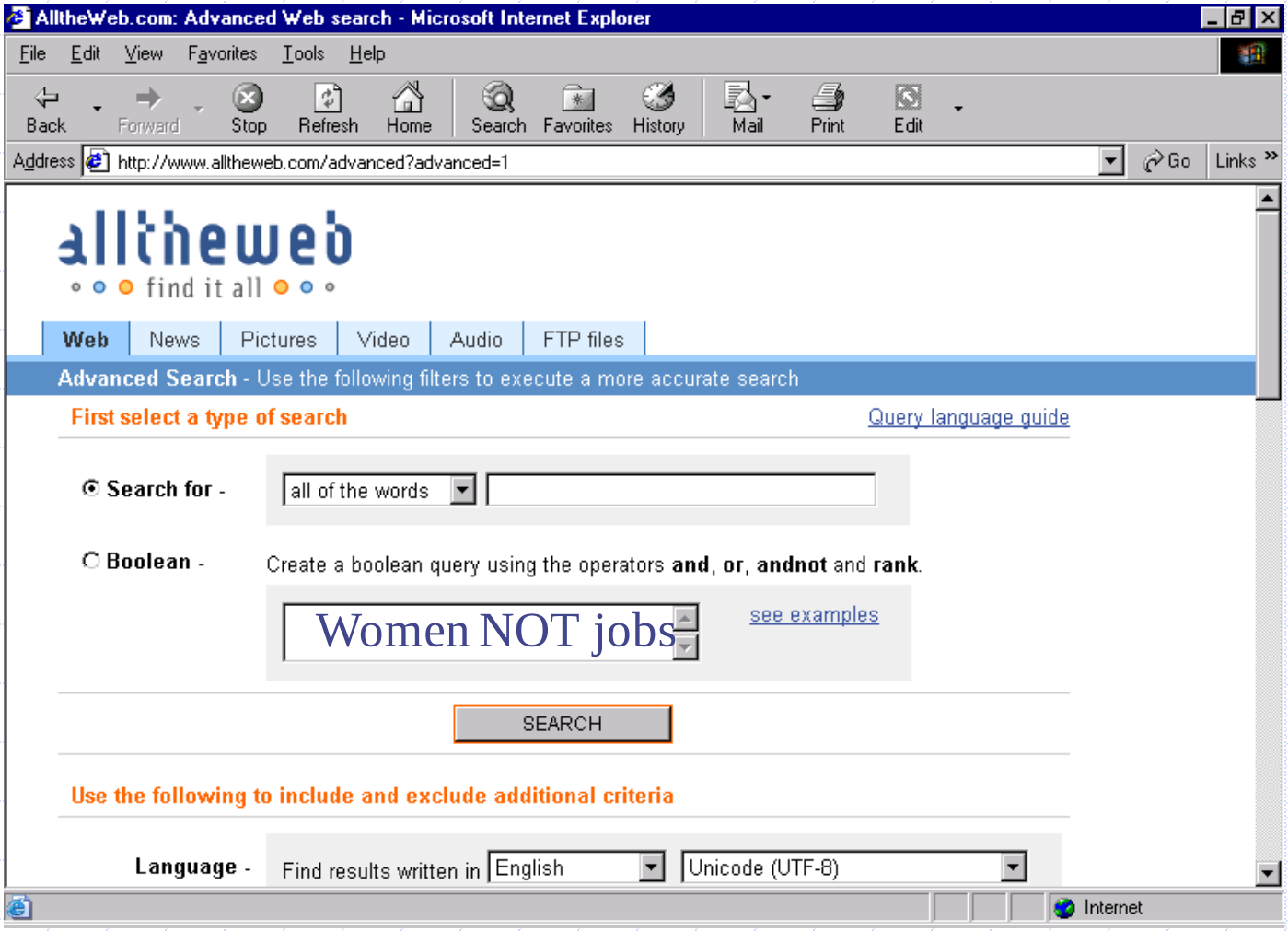
pink NOT floyd

pink -"pink floyd"




AlltheWeb.com: Advanced Web search - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Forward Stop Refresh Home Search Favorites History Mail Print Edit
Address http://www.alltheweb.com/advanced?advanced=1
Go Links
alltheweb
find it all
Web News Pictures Video Audio FTP files
Advanced Search - Use the following filters to execute a more accurate search
First select a type of search
Query language guide
Search for - all of the words
Boolean - Create a boolean query using the operators and, or, andnot and rank.
Women and jobs
see examples
SEARCH
Use the following to include and exclude additional criteria
Language - Find results written in English Unicode (UTF-8)
Internet

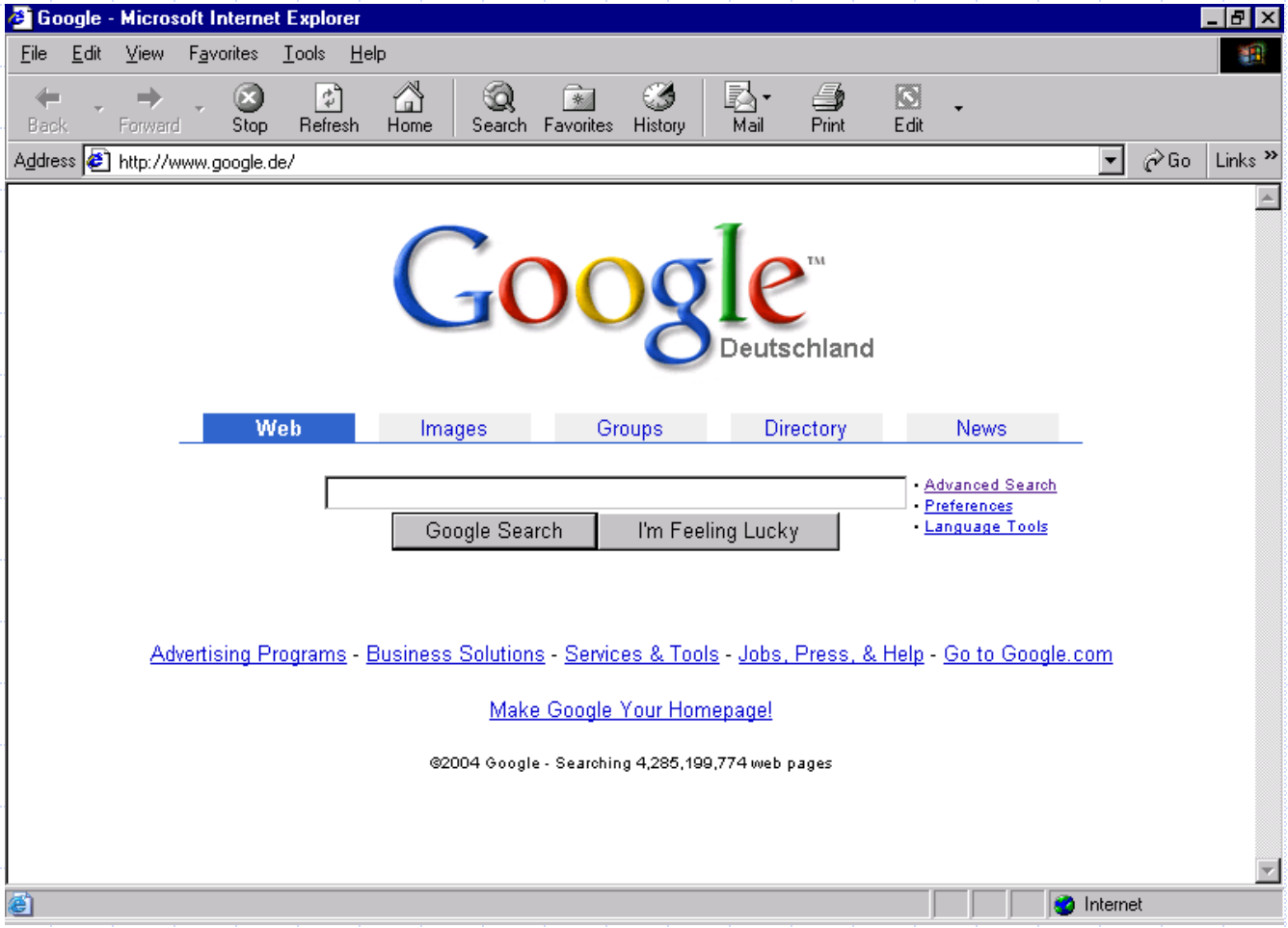

AlltheWeb.com: Advanced Web search - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Forward Stop Refresh Home Search Favorites History Mail Print Edit
Address http://www.alltheweb.com/advanced?advanced=1
Go Links
alltheweb
find it all
Web News Pictures Video Audio FTP files
Advanced Search - Use the following filters to execute a more accurate search
First select a type of search
Query language guide
Search for - all of the words
Boolean - Create a boolean query using the operators and, or, andnot and rank.
Women or jobs
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AlltheWeb.com: Advanced Web search - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Back Forward Stop Refresh Home Search Favorites History Mail Print Edit
Address http://www.alltheweb.com/advanced?advanced=1
Go Links
alltheweb
find it all
Web News Pictures Video Audio FTP files
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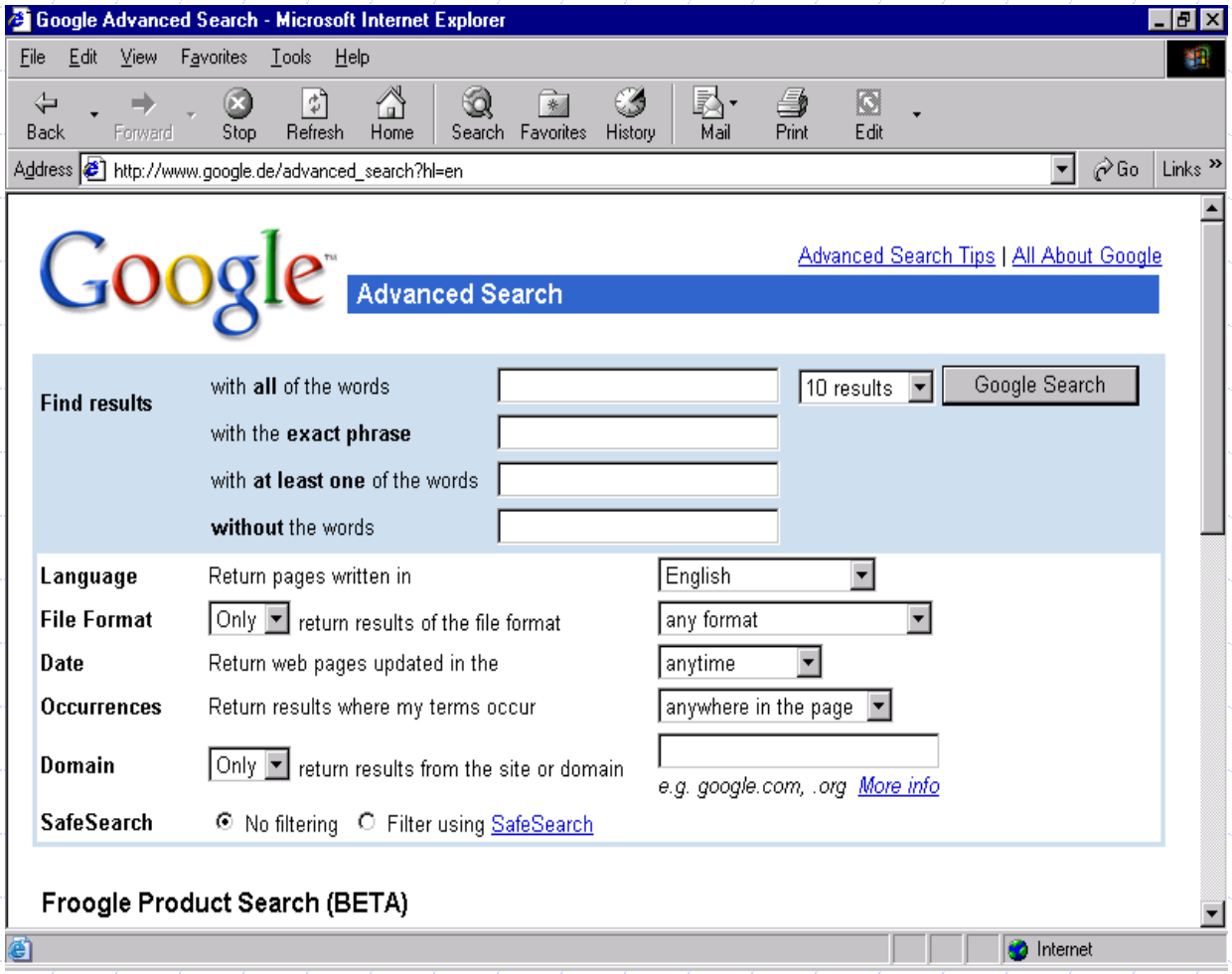
Combining Boolean Operators

- Boolean Operators can be combined e.g. (Women or jobs) and Malawi.
- You can use parenthesis or combine directly.
- Helpful when you are not sure of the spelling e.g. (Sarajevo or Sarayevo) and human rights.

The Advanced Search Option



Gives you a number of ways to limit your search



Get a second opinion

- Statistics say no search engine has it all:
  - Only about 60% of pages in Google are also in other search engines
  - Only 50% of pages in any search engine database are also found in all others
  - www.searchengineshowdown.com/stats/
- Use another large search engine

CRITICAL EVALUATION
Why Evaluate What You Find on the Web?

- Anyone can put up a Web page
  - about anything
  - for pennies
- in minutes
- Many pages not kept up-to-date
- No quality control
  - most sites not “peer-reviewed”
    - less trustworthy than scholarly publications
  - no selection guidelines for search engines